



UNIVERSITY^{AT}ALBANY
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Spring 2026)

Machine Learning

Lecture 19: Advanced Topic - Decision Making

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Machine learning overview

- Supervised learning
 - Tasks: classification, regression
 - Techniques: regularization, error decomposition, kernel, ensemble, neural network
- Unsupervised learning
 - Clustering, dimension reduction
- Some many advanced topics:
 - More tasks: ranking
 - More theoretical understanding: statistical learning theory
 - More settings: reinforcement learning
 - More applications: computer vision, natural language processing, speech recognition
 - More perspective: rule learning
 - More trustworthy: explainable ML, privacy / fairness issues in ML
 - ...

Today

- Decision making-based machine learning
 - Active learning
 - Disagreement-based algorithm
 - Confidence-based algorithm
 - Clustering-based algorithm
 - Bandits
 - Reinforcement learning

Today

- Decision making in machine learning
 - Active learning
 - Disagreement-based algorithm
 - Confidence-based algorithm
 - Clustering-based algorithm
 - Bandits
 - Reinforcement learning

Binary image classification tasks



Dog or mop?

Binary image classification tasks



Dog or croissant?

Binary image classification tasks



Dog or bagel?

Binary image classification tasks



Dog or muffin?

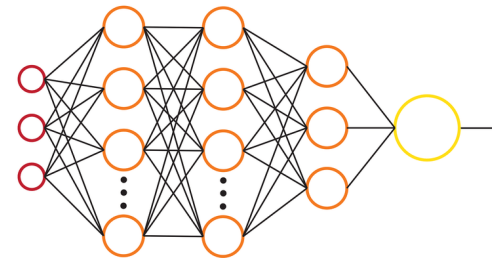
Binary image classification tasks



Dog or fried chicken?

Recap: Supervised learning

- How to achieve good performance?
 - Advanced techniques: kernel, ensemble, neural networks ...
 - Input **a lot of labeled images** of dogs or fried chicken

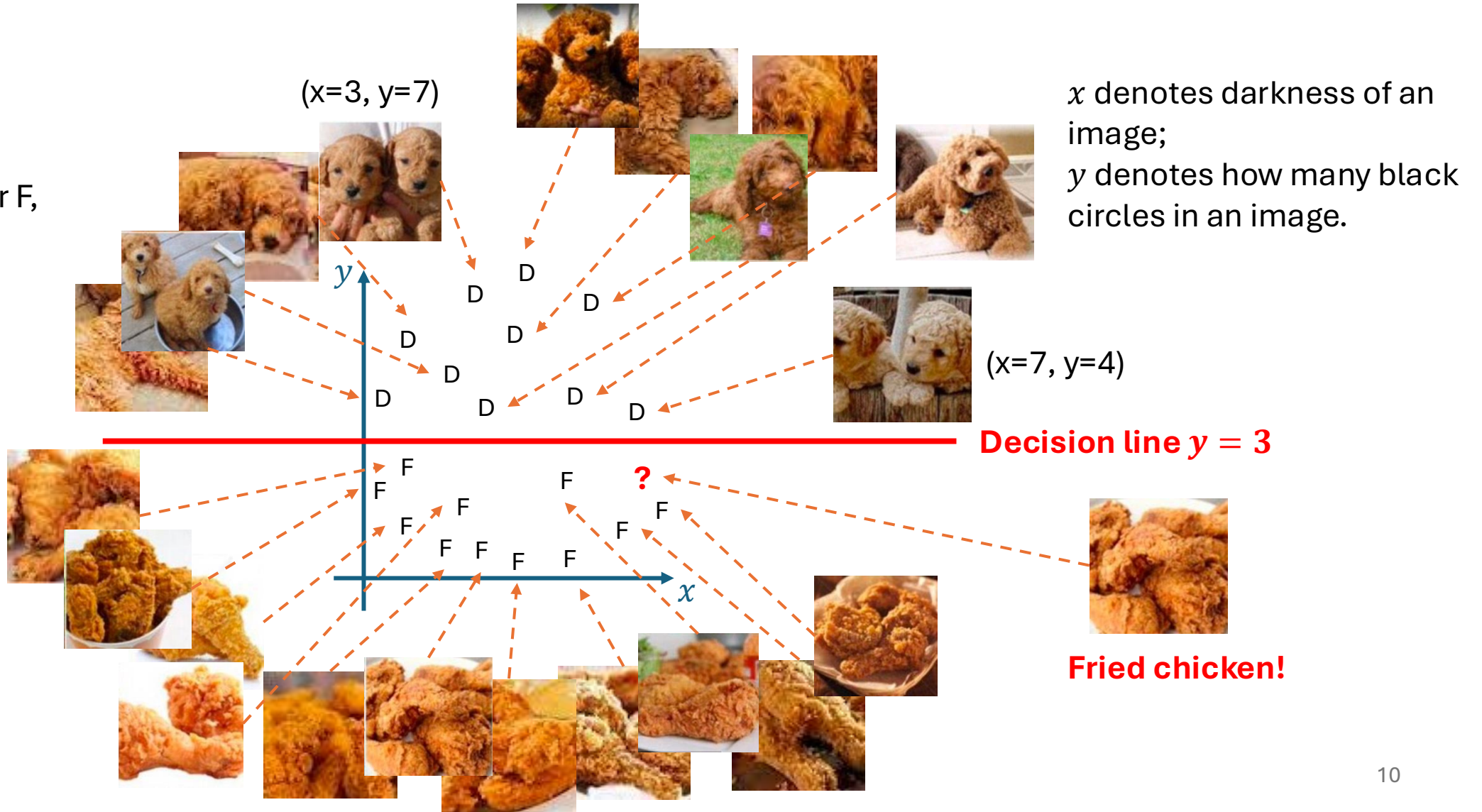


Given a new image, dog or fried chicken?



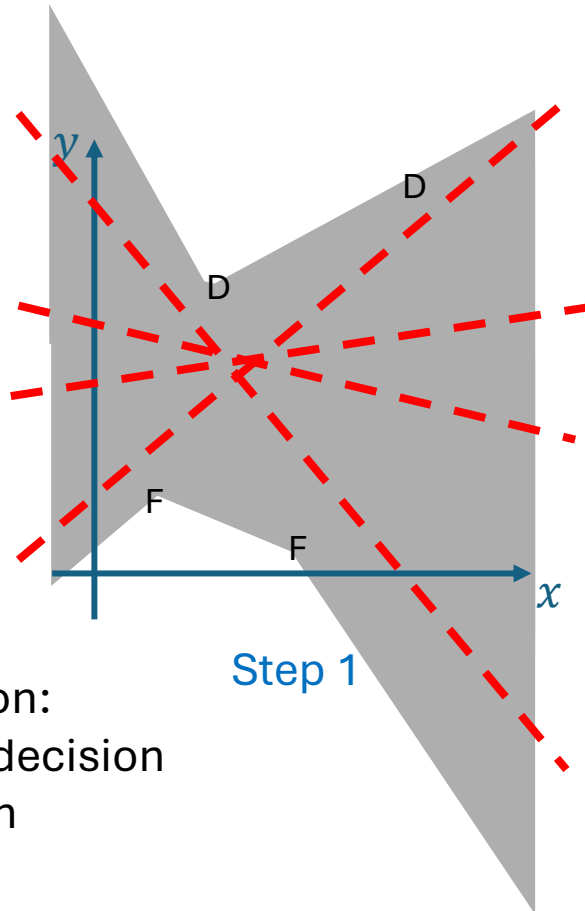
Binary classification in 2-dimension

Each point, D or F, represents an image.

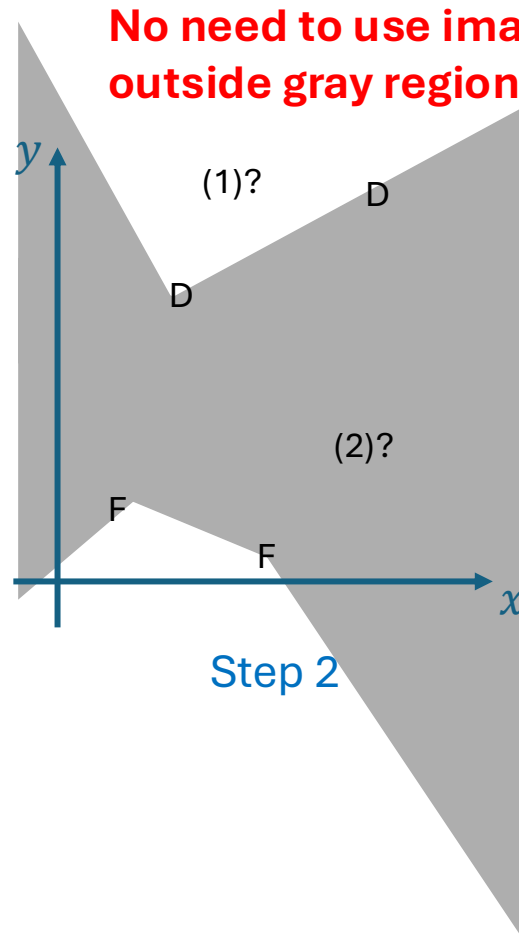


Fewer images to learn the decision line?

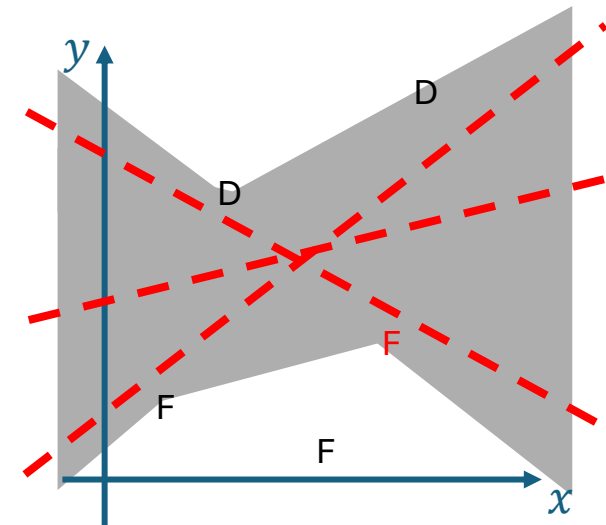
- Yes! Active learning selects important images.



Gray region:
Feasible decision
line region

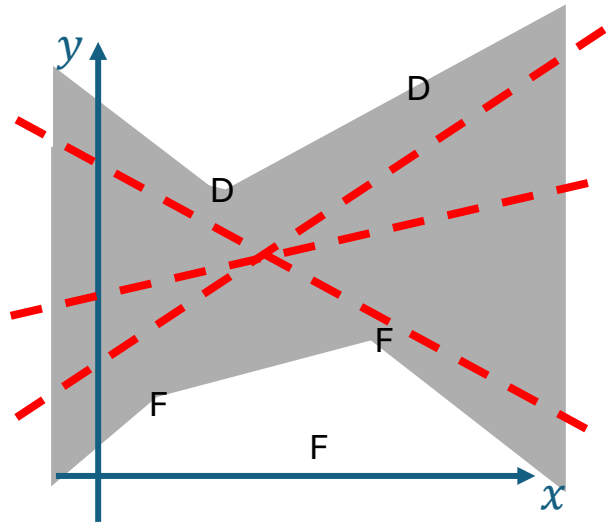


No need to use images
outside gray region.

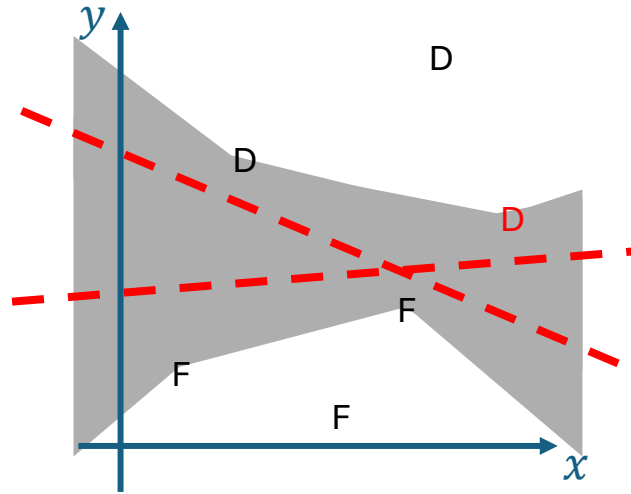


x denotes darkness of an image;
 y denotes how many black
circles in an image.

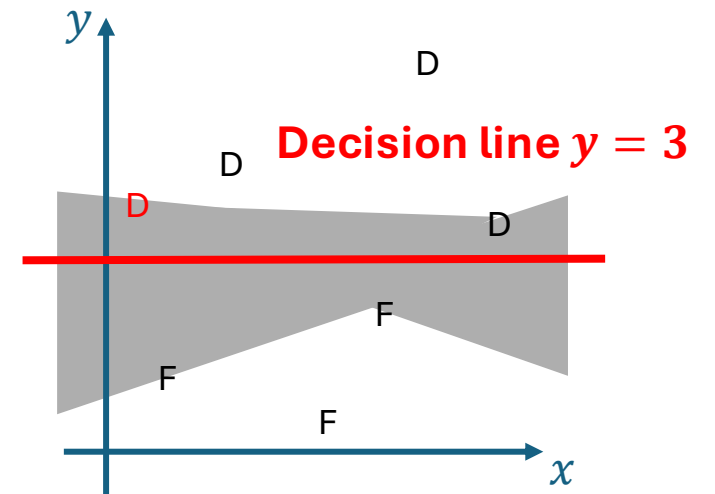
Moving forward ...



Step 3



Step 4



Step 5

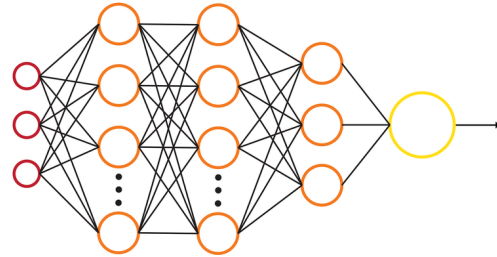
- Learn decision line with only 7 images!
 - Active learning only selects important images
 - images in gray region

Disagreement-based Active Learning

Confidence-based active learning



Labeled data

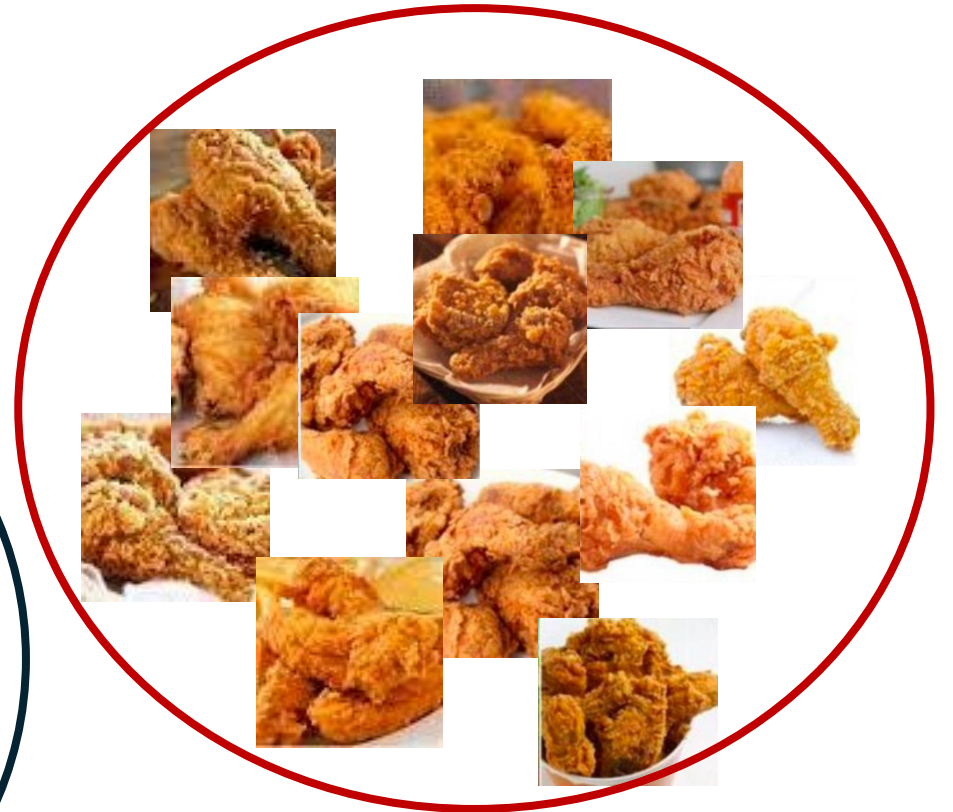


Unlabeled data

How to select the data point with least prediction confidence?

Clustering-based active learning

- Run k-means clustering
- Select samples from each cluster



Summary: active learning

- Framework:
 - Input: a small labeled dataset, a large unlabeled dataset
 - Output: a well-trained model
 - For $t = 1, 2, \dots$
 - Select a sample x_t from the unlabeled dataset
 - Get its label y_t
 - Update the model by adding (x_t, y_t)
- Goal:
 - Achieve good performance while **saving labeling cost** at the same time
- Can be super useful in practice!

Today

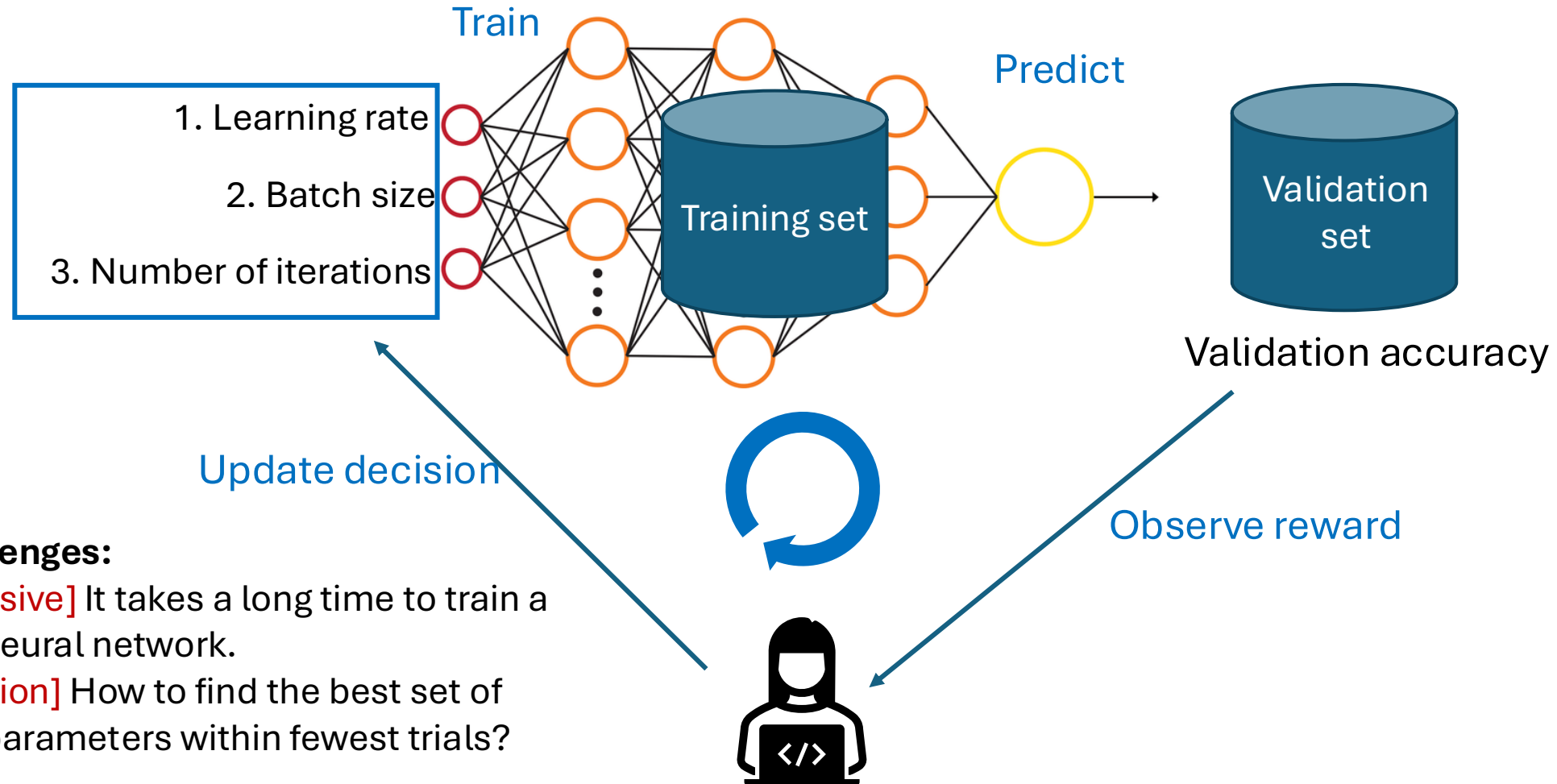
- Decision making in machine learning
 - Active learning
 - Disagreement-based algorithm
 - Confidence-based algorithm
 - Clustering-based algorithm
 - **Bandits**
 - Reinforcement learning

The real bandit machine

- Action
- Reward
- Goal:
 - maximize your reward
- Challenge:
 - You don't know what's behind the bandit machine



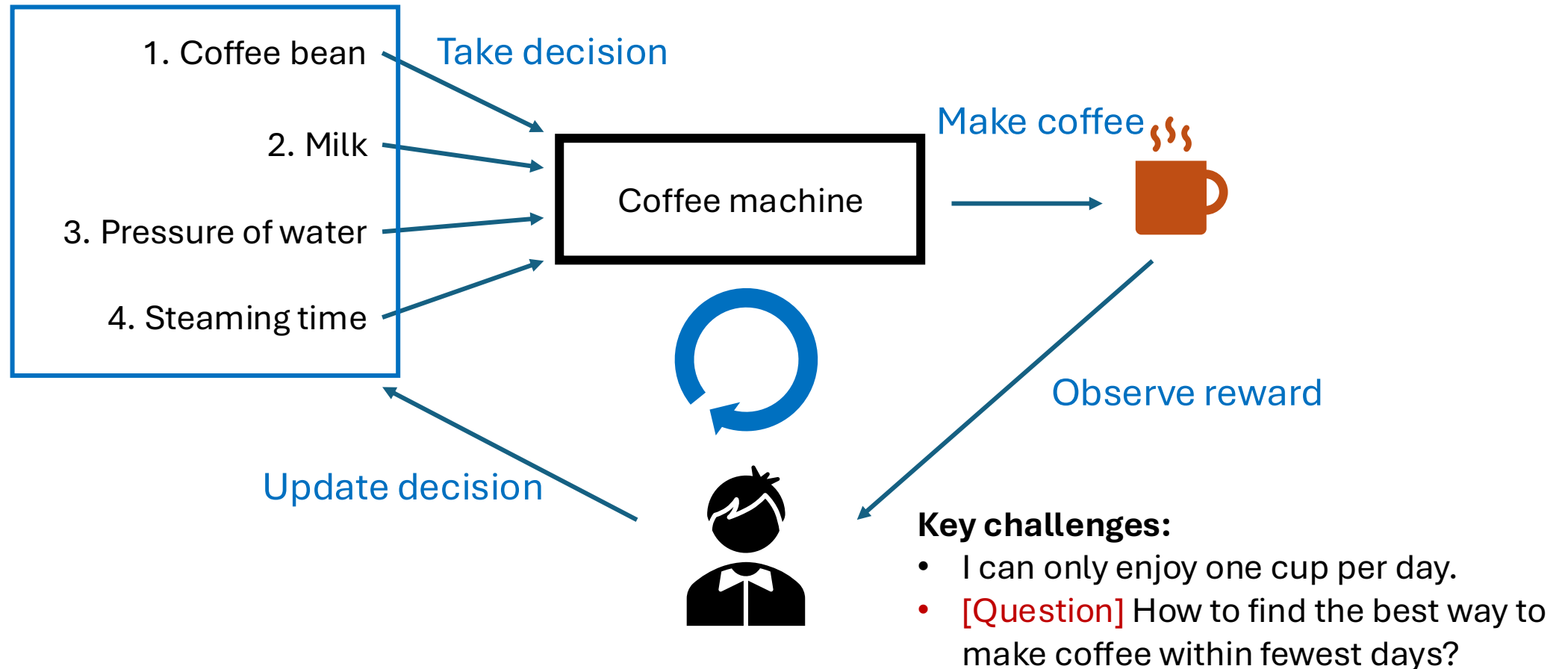
Bandits in hyperparameter tuning



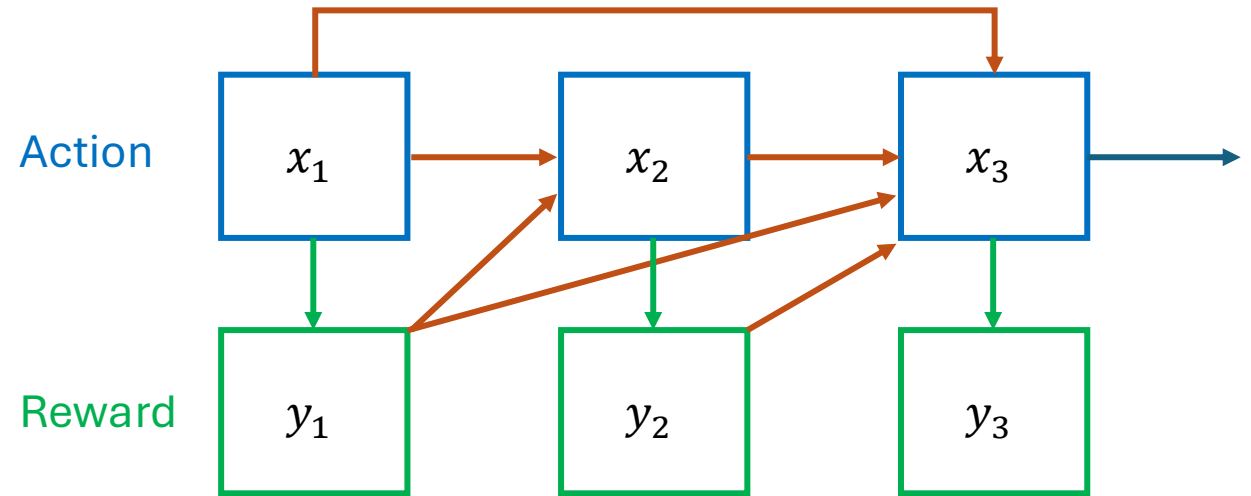
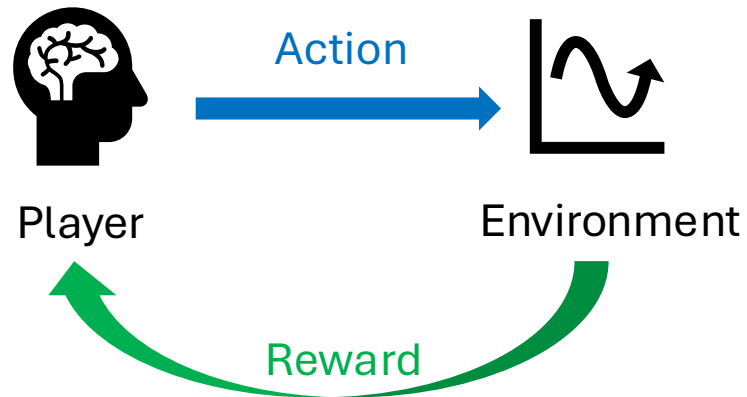
Key challenges:

- **[Expensive]** It takes a long time to train a deep neural network.
- **[Question]** How to find the best set of hyperparameters within fewest trials?

Bandits in making a cup of latte ...



Bandits framework

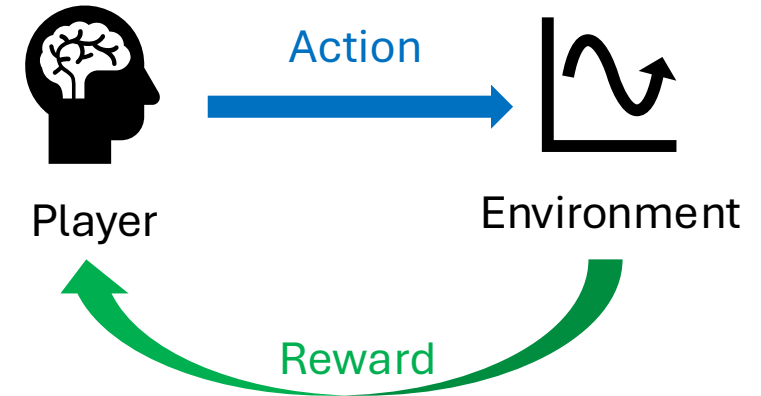


- [Example] hyperparameter tuning
 - x_1 : [learning_rate = 0.01, batch_size = 24, n_iterations = 200]
 - y_1 : validation_accuracy = 89%
 - x_2 : [learning_rate = 0.001, batch_size = 20, n_iterations = 250]
 - y_2 : validation_accuracy = 92%
 - ...

Two key challenges of bandits

1. A lot of time / computational cost / human effort

- 14 days for ImageNet-1k training!



2. No results of unselected actions

- No [learning_rate = 0.01, batch_size = 24, n_iterations = 200]
 - No validation accuracy of it
- How can we take fewer actions to find the action that maximizes the reward?

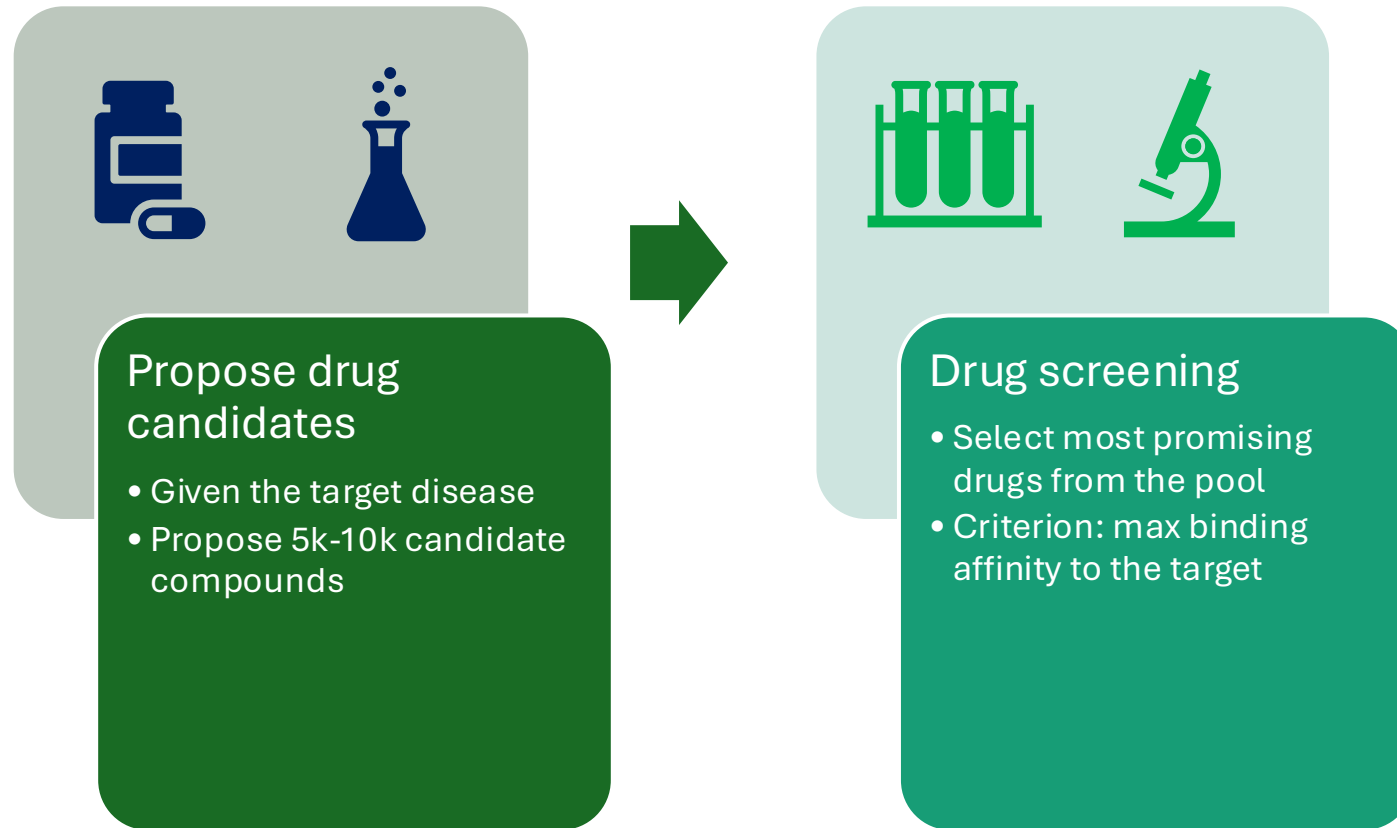
Expensive real-world drug discovery process



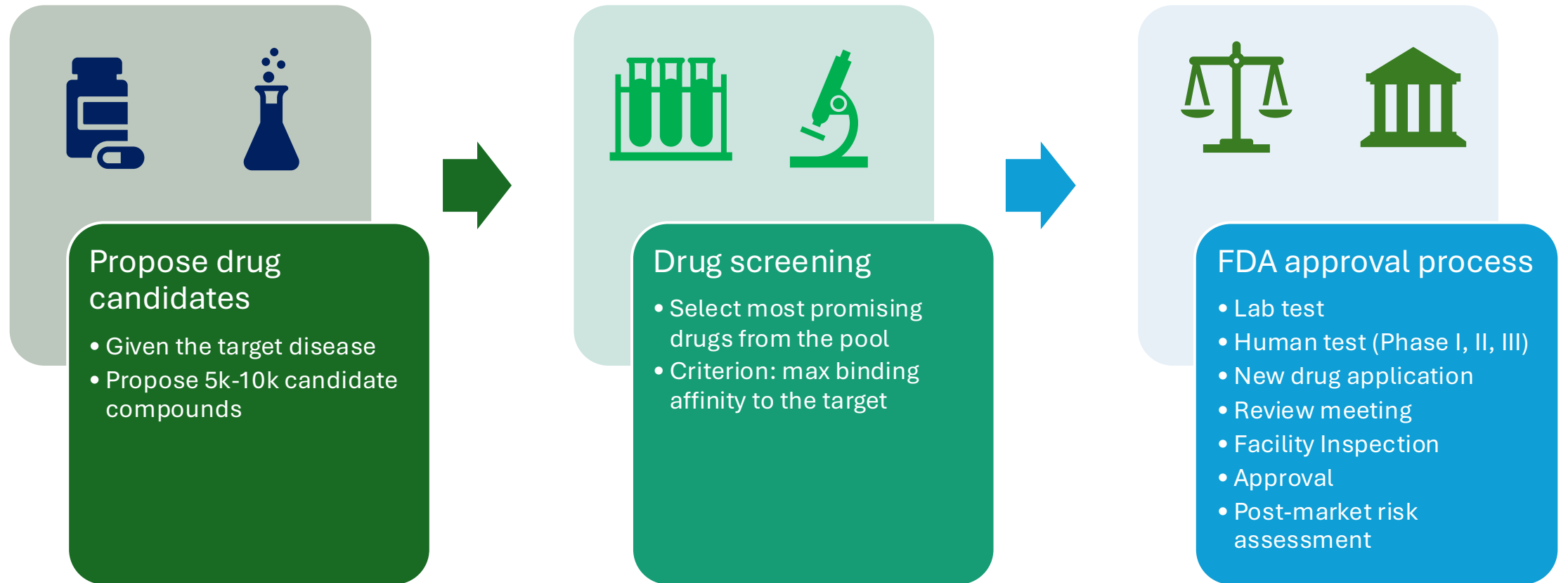
Propose drug candidates

- Given the target disease
- Propose 5k-10k candidate compounds

Expensive real-world drug discovery process



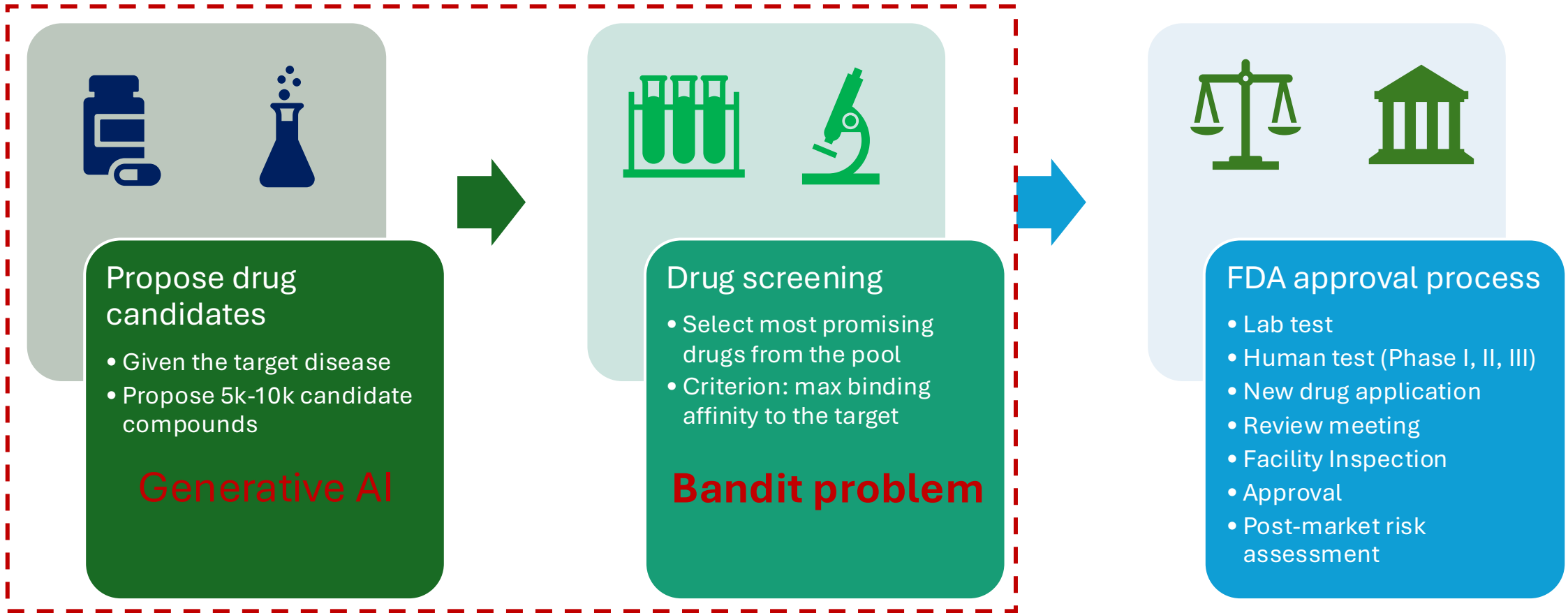
Expensive real-world drug discovery process



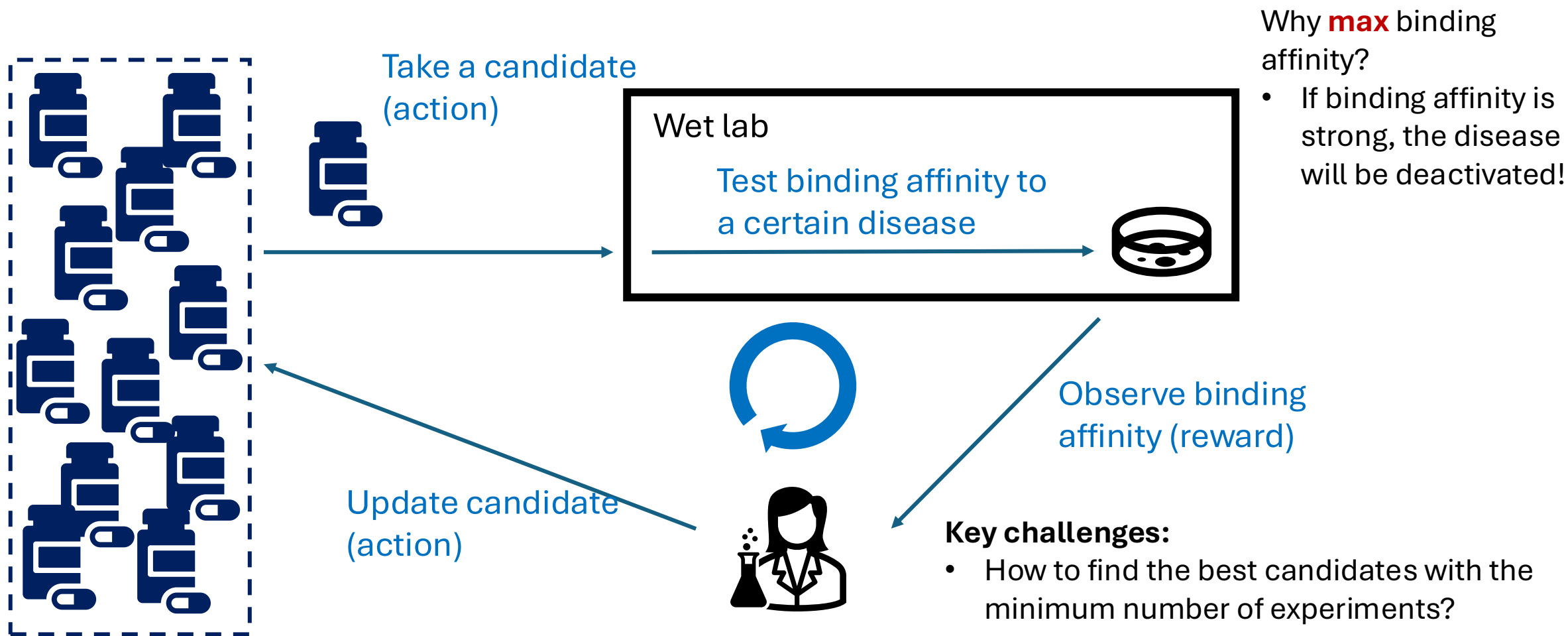
A new drug into market: **\$2.8 billion dollars** and **10-15 years** [Pushpakom et al., Nat Rev Drug Discov, 2018]

AI is now amazing. How can AI help?

Currently AI helps in Stage I & II.



Bandits in Stage II Drug Screening (Lab in the loop)



Key challenges of applying Bandits in drug screening

- Expensive wet lab evaluations in
 - **Time**
 - **Cost**
- How expensive?
 - Let's read some **latest** numbers.
 - Nov. 2025, Abtique, an online marketplace for molecule (drug candidate) validation

Press Releases

EVQLV Launches Abtique, the First Marketplace for AI-Designed Molecules Validated in Independent Labs

November 20, 2025 | 1 min read



New “click-to-sequence” platform lets researchers buy AI-designed antibody sequences online, using an AI system that learns from real lab results

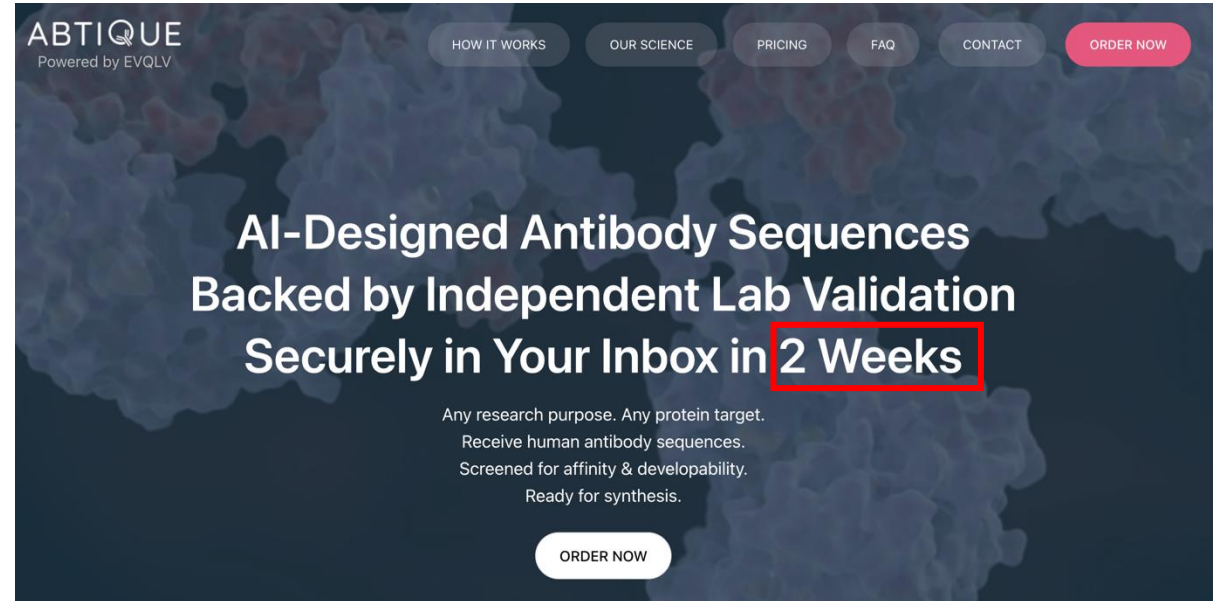
NEW YORK, Nov. 12, 2025 – EVQLV announced the launch of Abtique, a first-of-its-kind online marketplace, where scientists can order AI-designed antibody sequences in minutes. The debut coincides with the release of HyperBind2, EVQLV’s AI system for antibody design, which has been independently validated in dozens of labs and detailed in a new research paper.

Antibodies power many of the world’s leading medicines for cancer, arthritis, asthma, and more. The antibody market reached **\$246 billion in 2024** and is projected to grow to **\$685 billion by 2034** (Towards Healthcare, 2025). As scientists, researchers, and drug developers seek faster, less expensive ways to develop drugs, AI is advancing significantly across the life sciences. Yet building and validating these technologies requires years of work, significant costs, and rare expertise.

“Abtique answers the question: Why can’t every researcher working on groundbreaking science access AI-designed molecules?” said Brett Averso, CTO of EVQLV. “With Abtique, a scientist can order lab-ready antibody sequences in minutes, with no AI expertise required, backed by independent lab validation.”

Long time turnaround in drug validation

- Let's talk about time first
- Each evaluation: 2 weeks
 - If one sequentially works with bandit algorithm, one can test **only 26** drug candidates in **an entire year!**
 - Batched BO definitely helps, but how about money?



Huge cost in drug validation

- Most popular choice:
 - ~\$200 per drug candidate
- A successful Stage I - Drug Proposal
 - At least 5k-10k candidates
 - That's **\$1M-2M** dollars!

The screenshot displays the ABTIQUE website, which is powered by EVQLV. The navigation bar includes links for 'HOW IT WORKS', 'OUR SCIENCE', 'PRICING', 'FAQ', 'CONTACT', and a prominent 'ORDER NOW' button. The main heading states: 'One-time project pricing. No milestones. No royalties. No subscriptions.' Below this, it notes 'Transparent project-based pricing for every stage of discovery.'

Three pricing tiers are presented:

- DISCOVERY**: Priced at **\$9,600**. It is 'Designed for early exploration' and 'For small-scale or early discovery projects'. It includes 24 target-specific sequences, a single discovery round for rapid feasibility testing, diversity across CDRs for targeted insight, and non-exclusive sequences.
- TARGETED** (Most Popular): Priced at **\$18,600** (reduced from \$38,400). It is 'Designed for hit discovery' and 'For scientists expanding design confidence'. It includes 96 target-specific sequences, a single discovery round for efficient lead identification, broader diversity across CDRs for early expansion, non-exclusive sequences, and suitability for early-stage hit/lead generation.
- EXCLUSIVE**: Priced at **\$48,600** (reduced from \$159,200). It is 'Validated multi-round refinement' and 'For scientists advancing iterative discovery and sequence protection'. It includes 288 target-specific sequences (over 3 rounds), three feedback-informed rounds of 96 sequences, HyperBind-guided refinement between each round, the broadest diversity across CDRs to explore full candidate space, EXCLUSIVITY on the project sequences, and validation in published results demonstrating multi-round refinement.

Each tier has an 'ORDER NOW' button at the bottom.

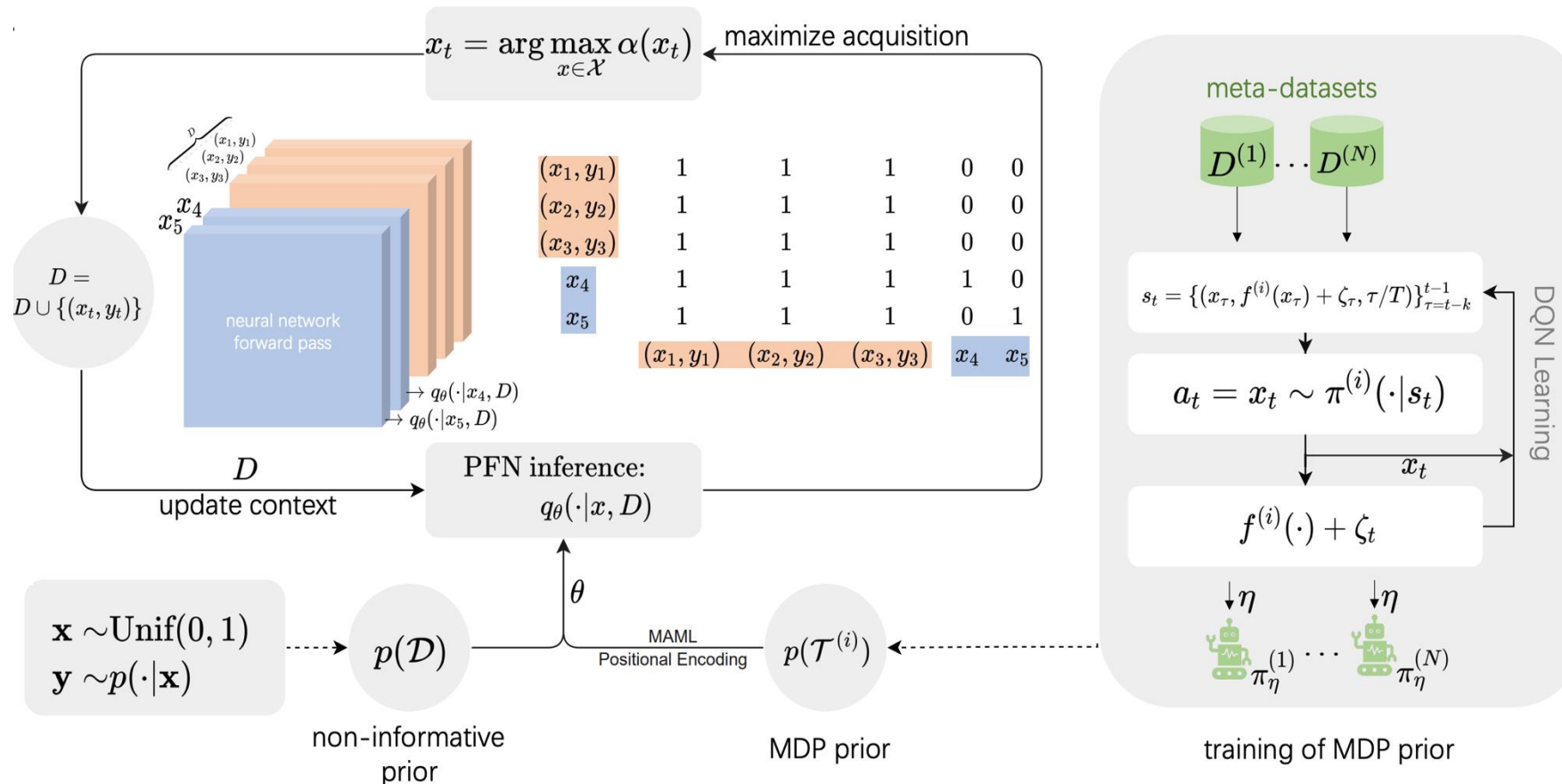
So, an urgent question for the entire drug discovery industry



*How can we design a bandit algorithm that finds **highly promising** drug candidates within very **few** rounds?*

How few? Just 20 rounds!

Recent work: **Procedure-informed BO (ProfBO)** framework



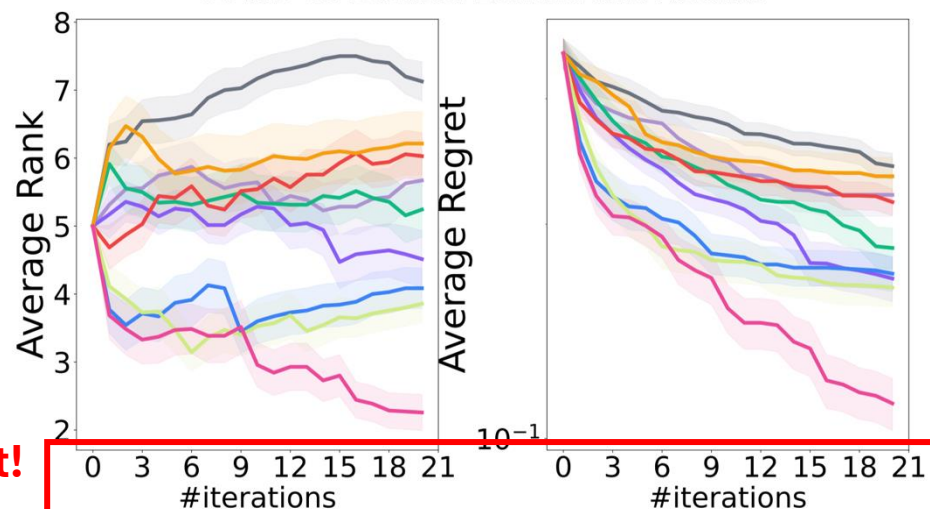
Small molecule binder discovery

- We search for small molecules that have high docking scores for Covid-19 [Liu et al., 2023; Sterling & Irwin, 2015] and Cancer receptors.
 - ProfBO achieves significantly **lower regret** and **better rank** than all baselines.
 - ProfBO rapidly identifies high-quality molecules in the **top 0.5% of all candidates**.

What does this mean?

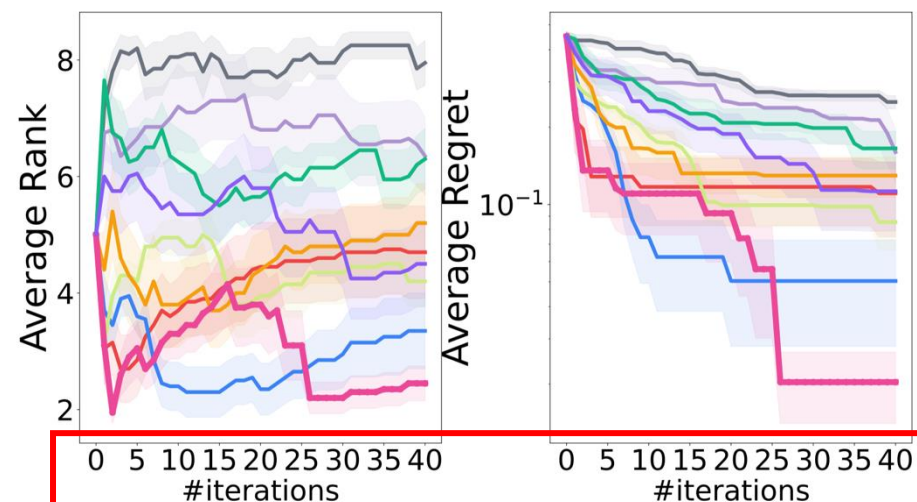
- E.g., in a **5k** drug candidate pool, ProfBO finishes screening within **25 rounds**
 - Saving evaluation time from 200 years (sequentially one-by-one) to **50 weeks!**

COVID-19 Dataset Benchmark Results



Few-shot!

Cancer Dataset Benchmark Results



Today

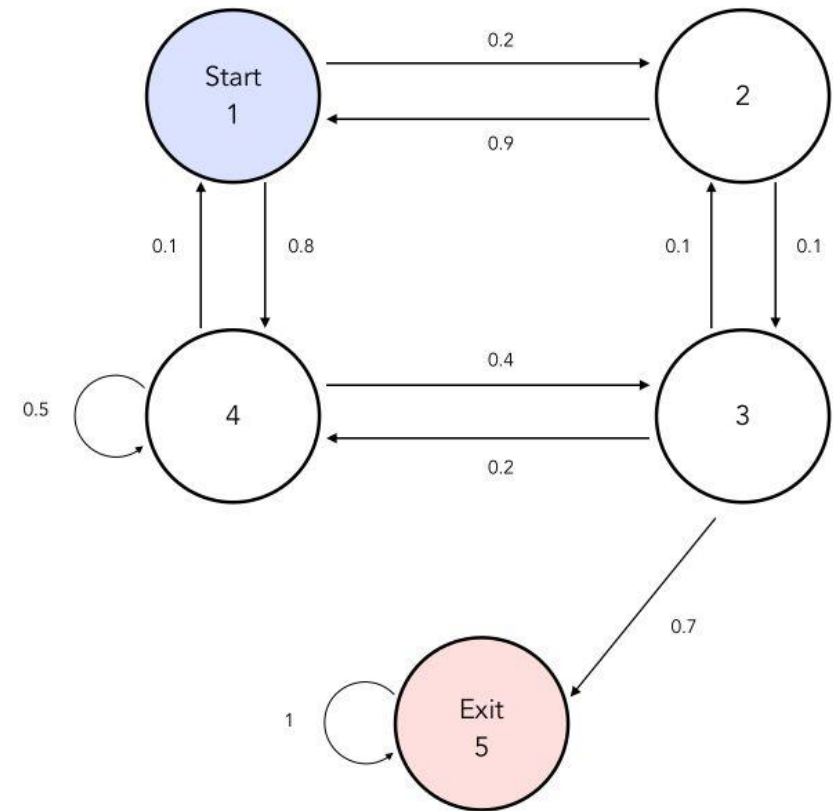
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Corn maze



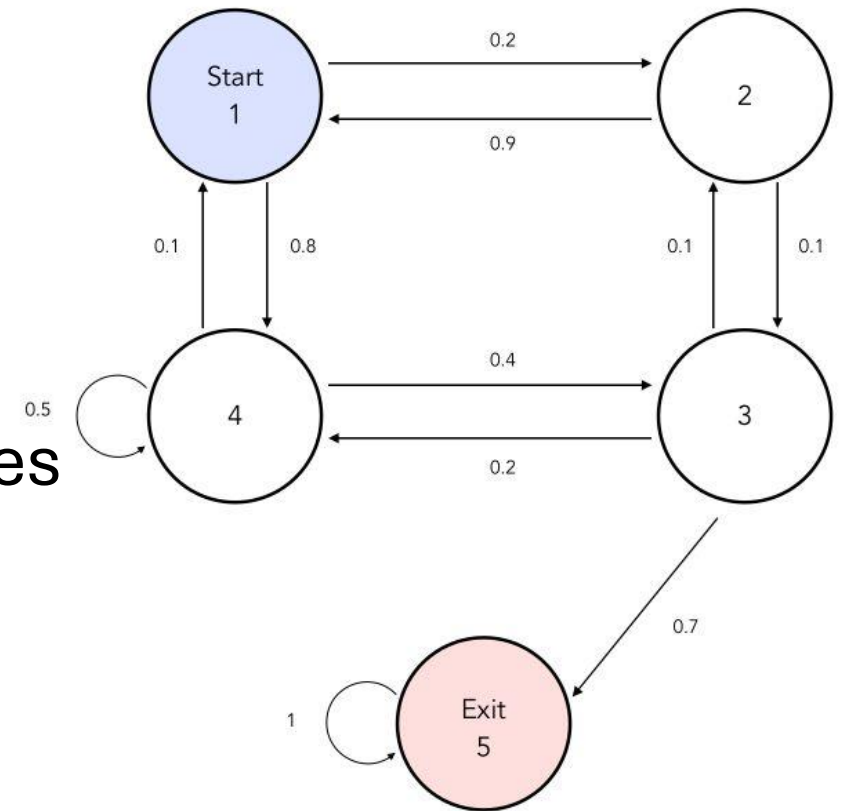
Markov decision process

- State: different locations
- Action: move
- Reward: exit
- Key design:
 - Once you are in a new state, the new state doesn't depend on historical states



Key challenges of reinforcement learning

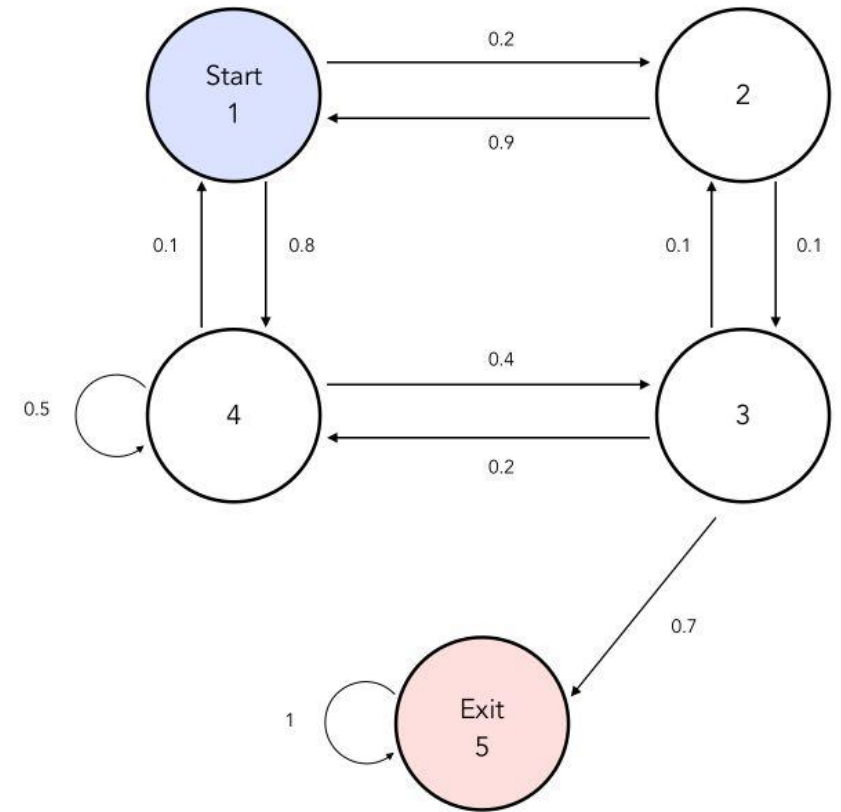
- Sample efficiency
- Delayed rewards
- High-dimensional state and action spaces
- Safety constraints



How to take action? Epsilon-greedy algorithm

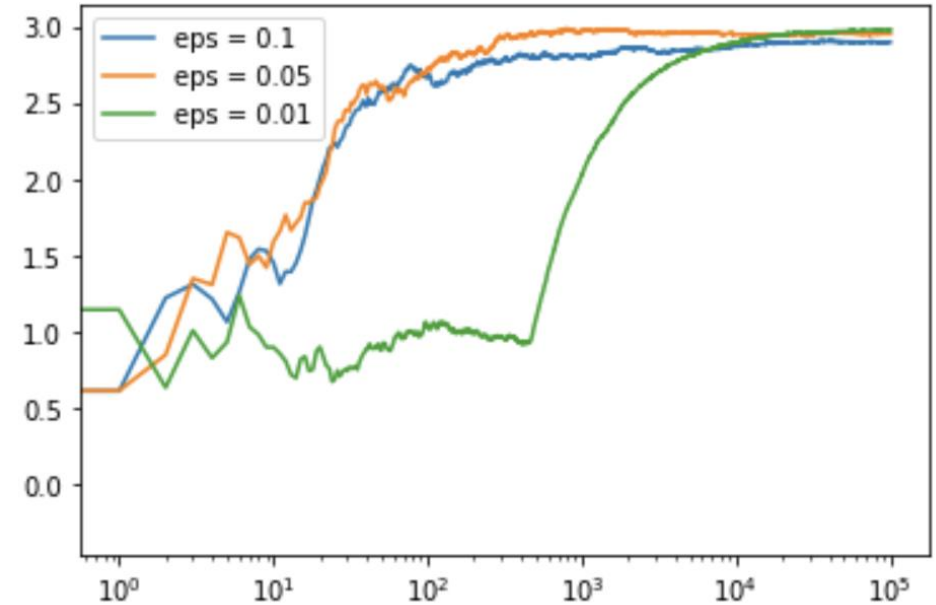
Action at time(t) $\left\{ \begin{array}{ll} \max Q_t(a) & \text{with probability } 1-\epsilon \\ \text{any action } (a) & \text{with probability } \epsilon \end{array} \right.$

- **Exploration-Exploitation trade-off in RL**
 - After playing with the environment, you have learned something
 - You know your current best action – you can keep taking that [Exploitation]
 - But there are so many actions that you don't know
 - You need to explore a bit [Exploration]



Epsilon-greedy algorithm

- Exploration-Exploitation trade-off in RL
 - After playing with the environment, you have learned something
 - You know your current best action – you can keep taking that [Exploitation]
 - But there are so many actions that you don't know
 - You need to explore a bit [Exploration]



More exploration – large epsilon
More exploitation – small epsilon

Summary: frameworks

- Active Learning

- Input: a labeled dataset, an unlabeled dataset
- Output: a well-trained model
- For $t = 1, 2, \dots$
 - Select a sample x_t from the unlabeled dataset
 - Get its label y_t
 - Update the model by adding (x_t, y_t)

- Goal: learning a good mapping $X \rightarrow Y$

- Bandits

- Input: an open environment
- Output: a policy
- For $t = 1, 2, \dots$
 - Take action x_t according to decision model
 - Observe its reward y_t
 - Update the decision model by adding (x_t, y_t)

- Goal: learning a good policy to take action

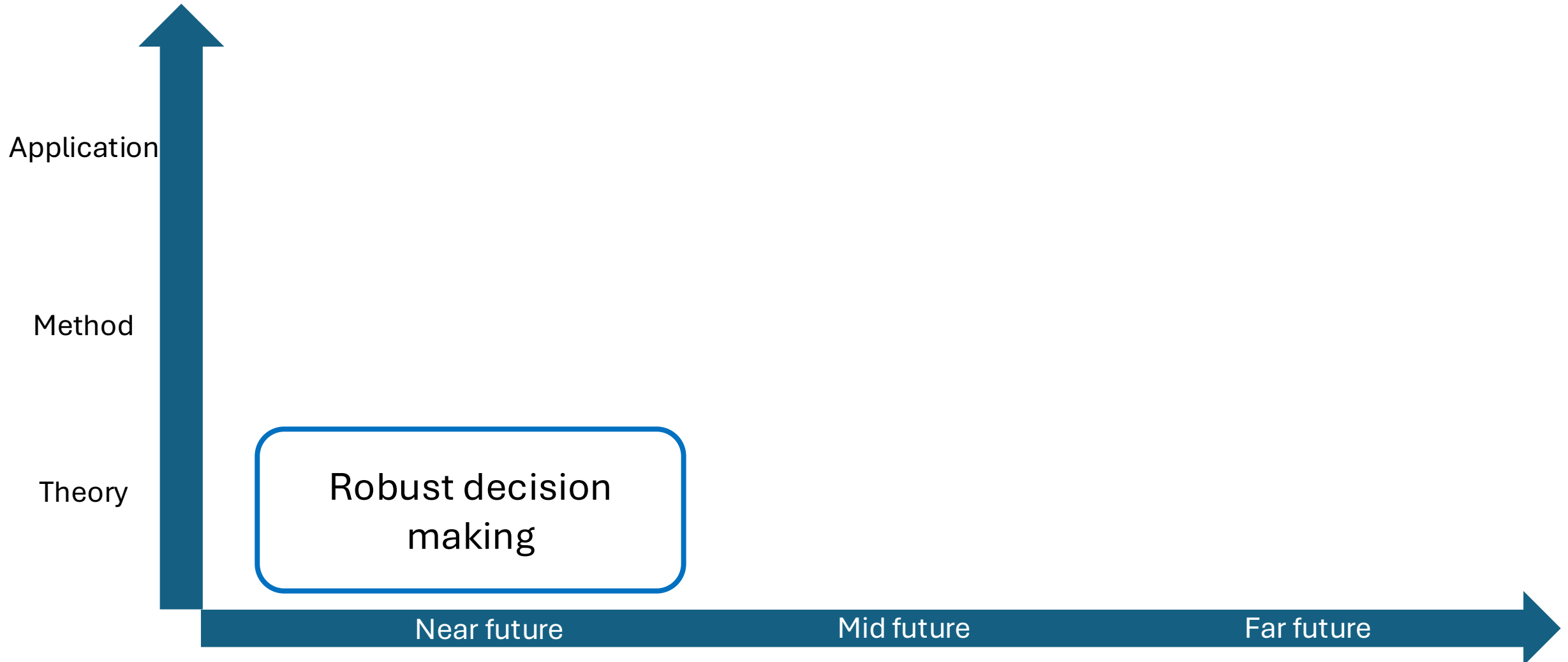
- Reinforcement Learning

- Input: an open environment
- Output: a policy
- For $t = 1, 2, \dots$
 - Take action x_t according to decision model
 - Move to state s_t and observe its reward y_t
 - Update the decision model by adding (x_t, s_t, y_t)

- Goal: learning a good policy to take action given state

Future vision of decision making

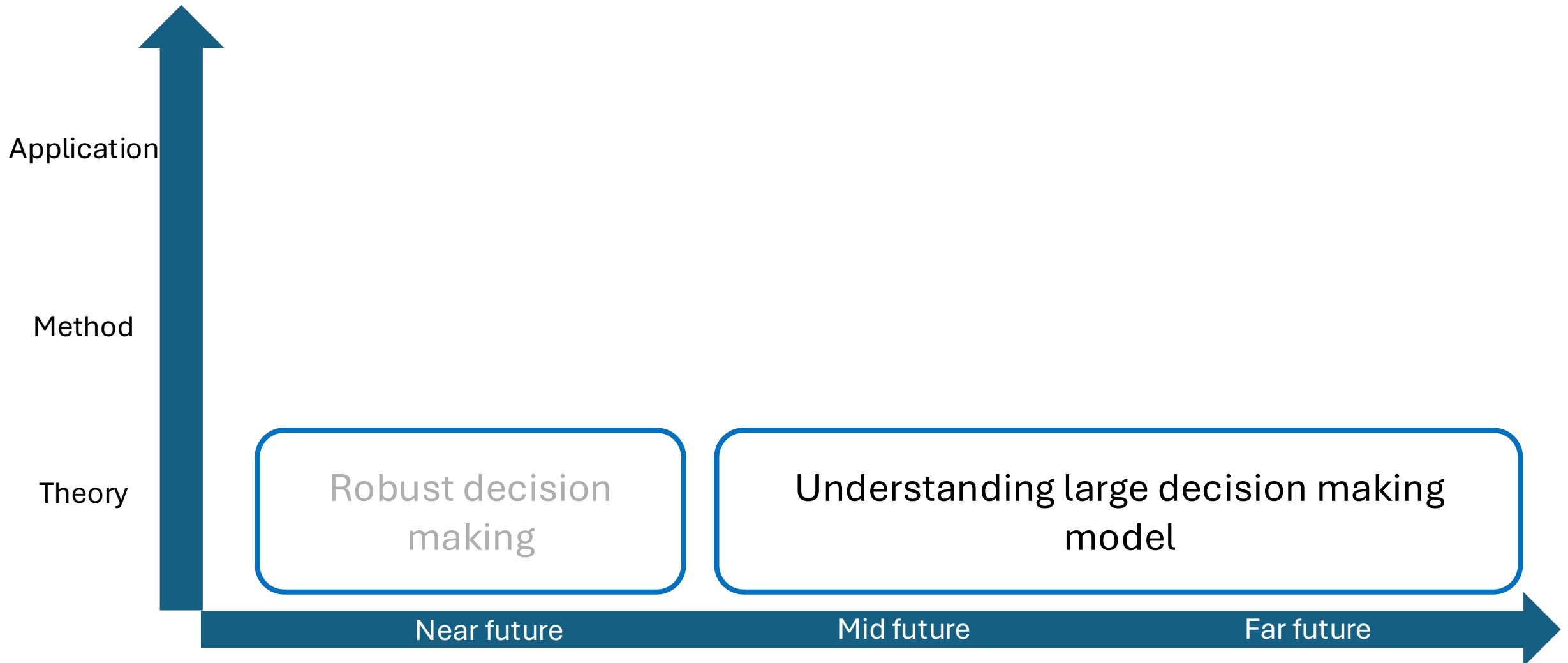
Future vision of decision making



Robust decision making

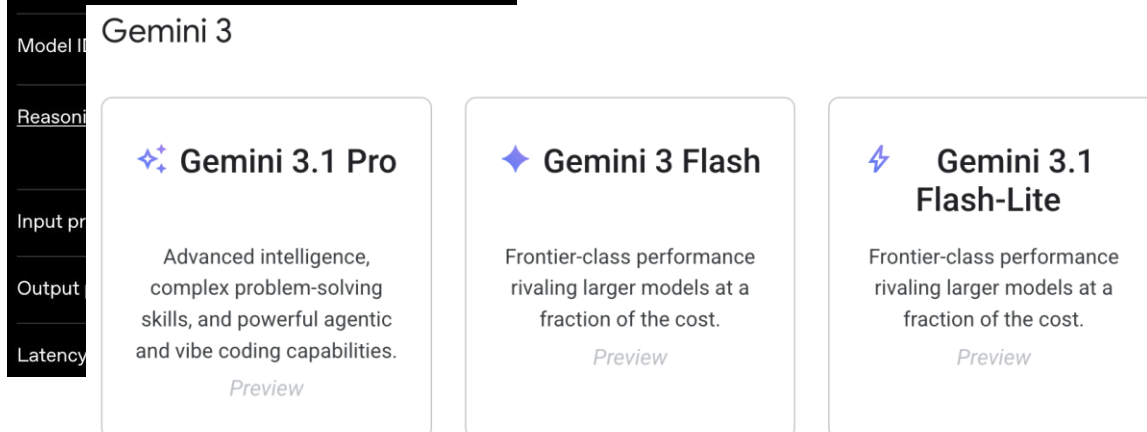
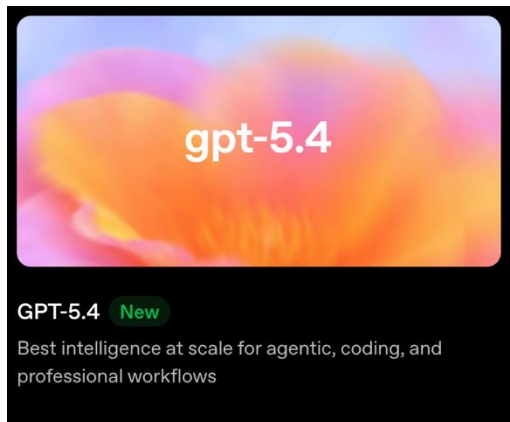
- How to make decisions in more challenging settings?
 - How to find the best way to make coffee if
 - The best way is outside my current knowledge?
- Non-linear bandits?
- Kernelized bandits?
- Reinforcement learning?

Future vision of decision making



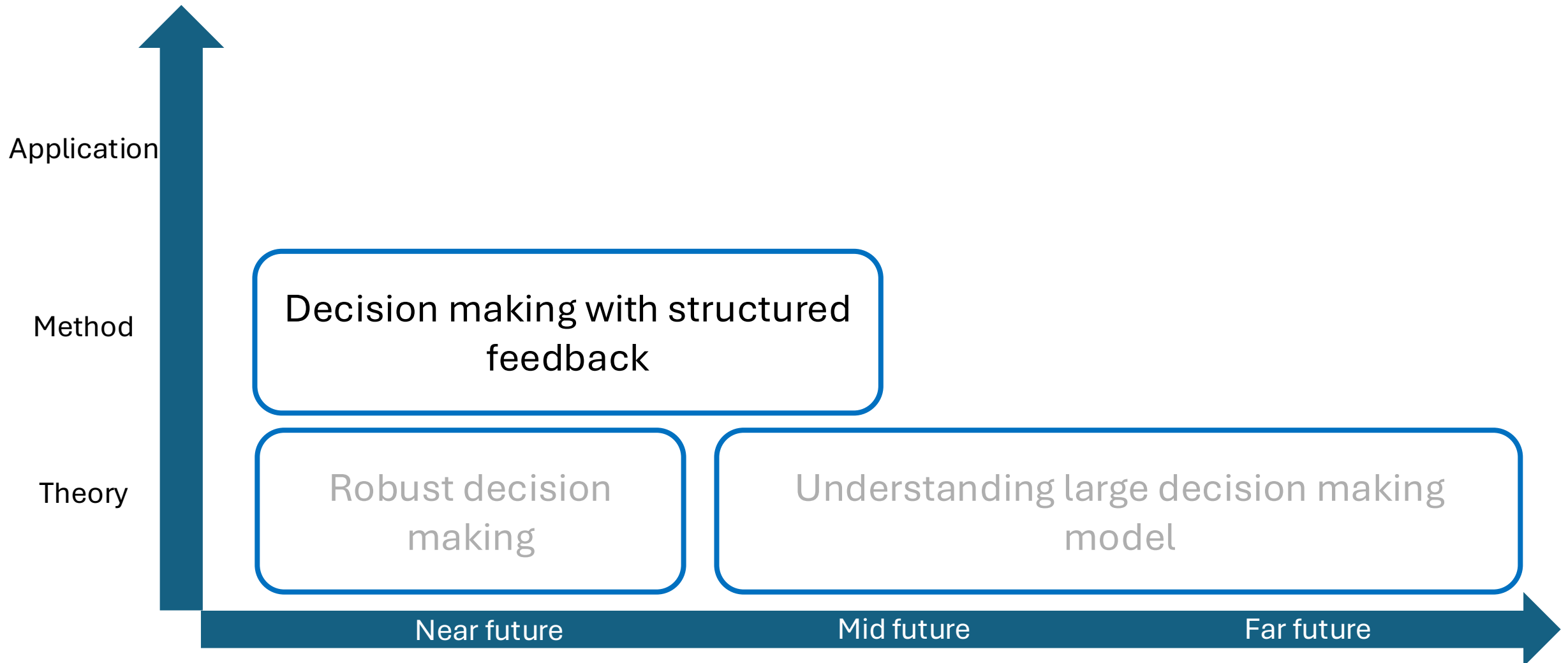
Understanding large decision making model

- Large model stronger than ever
- Powerful but also dangerous



- Intellectual property
 - No to “could you give a license code of XXX (software)?”
- Privacy
 - No to “could you tell more about YYY (person)?”

Future vision of decision making



Decision making with structured feedback

1. Pairwise feedback

- Application:
 - Hardness of materials

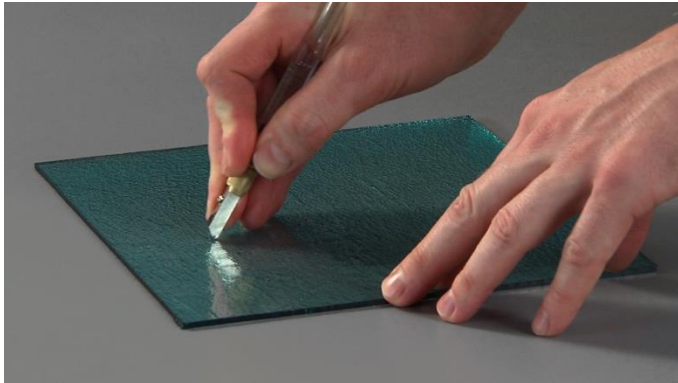
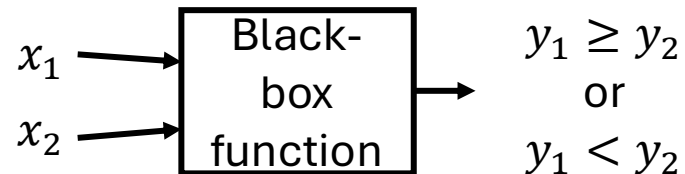
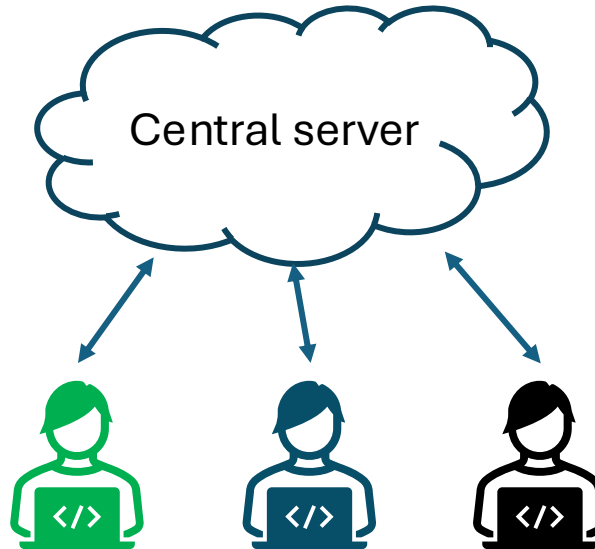


Figure from Bullseye Glass



2. Federated feedback

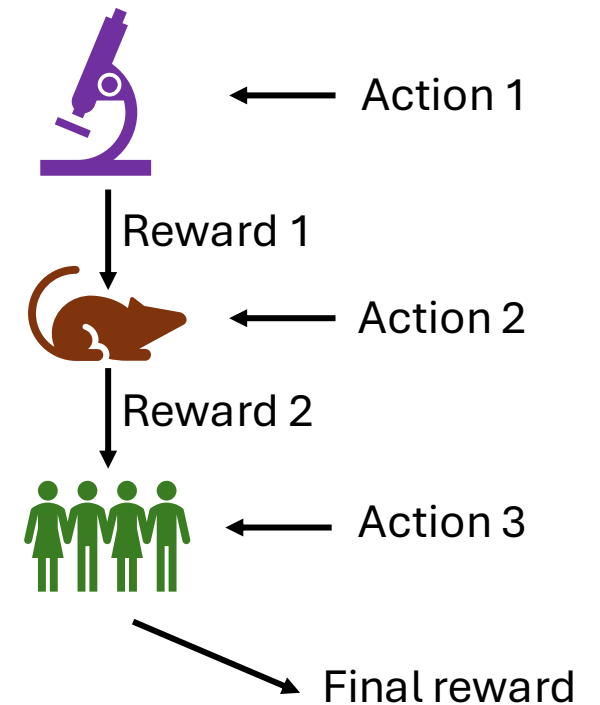
- Application:
 - Private computing
 - Large-scale computing



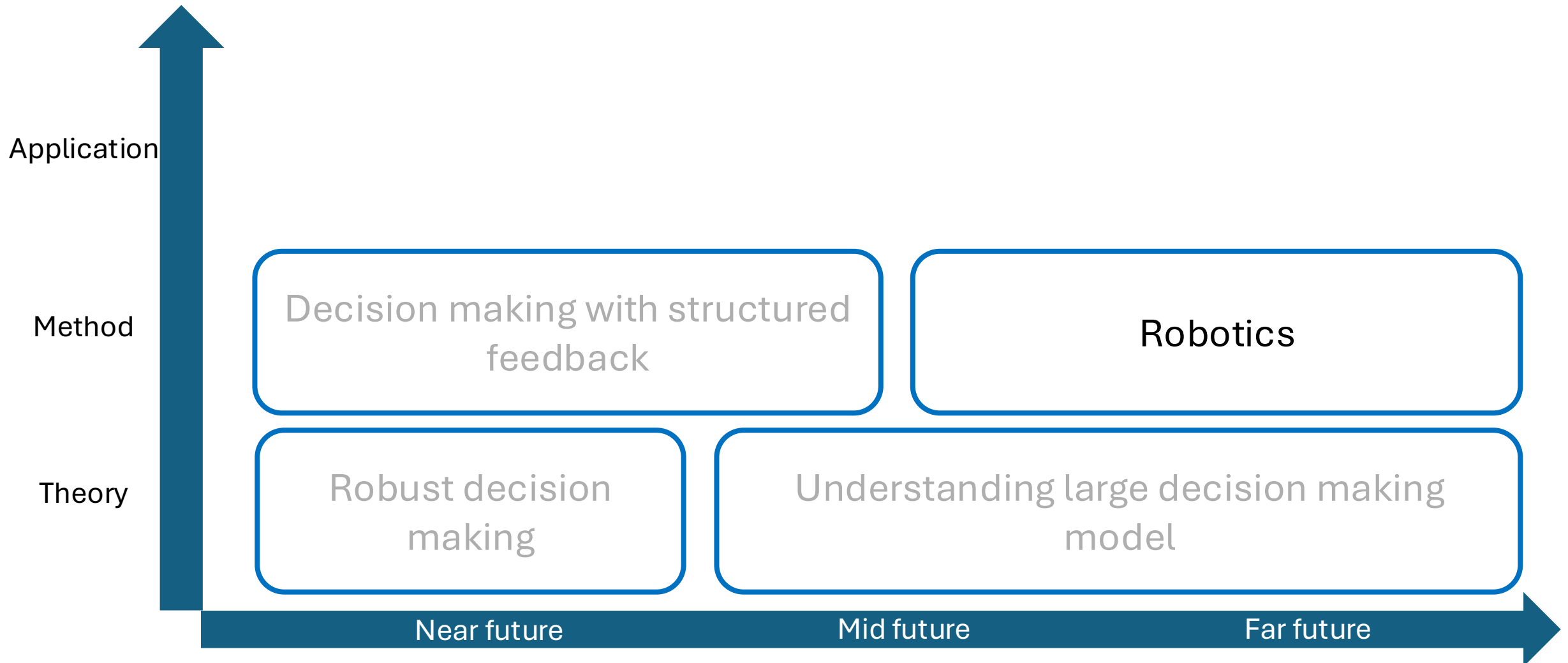
- Separate actions and reward

3. Multi-step feedback

- Application:
 - 3 phases in vaccine development



Future vision of decision making

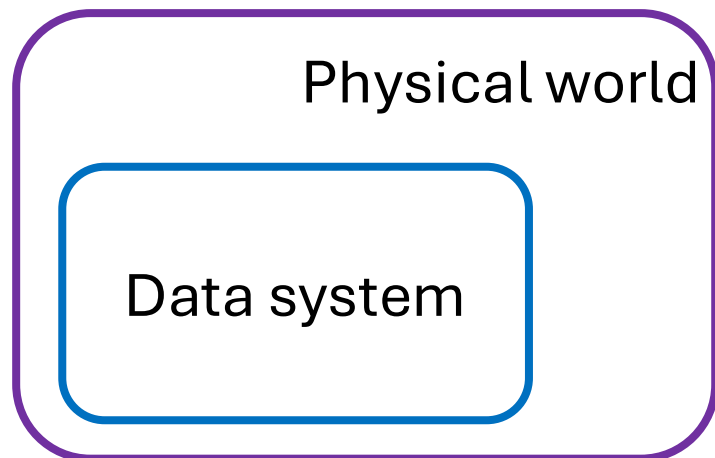


Robotics: decision making in physical world

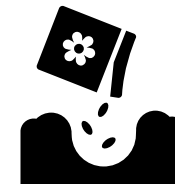
- Data-to-data decision making will be perfect in the future



- Decision making in physical world is truly needed



Car repair

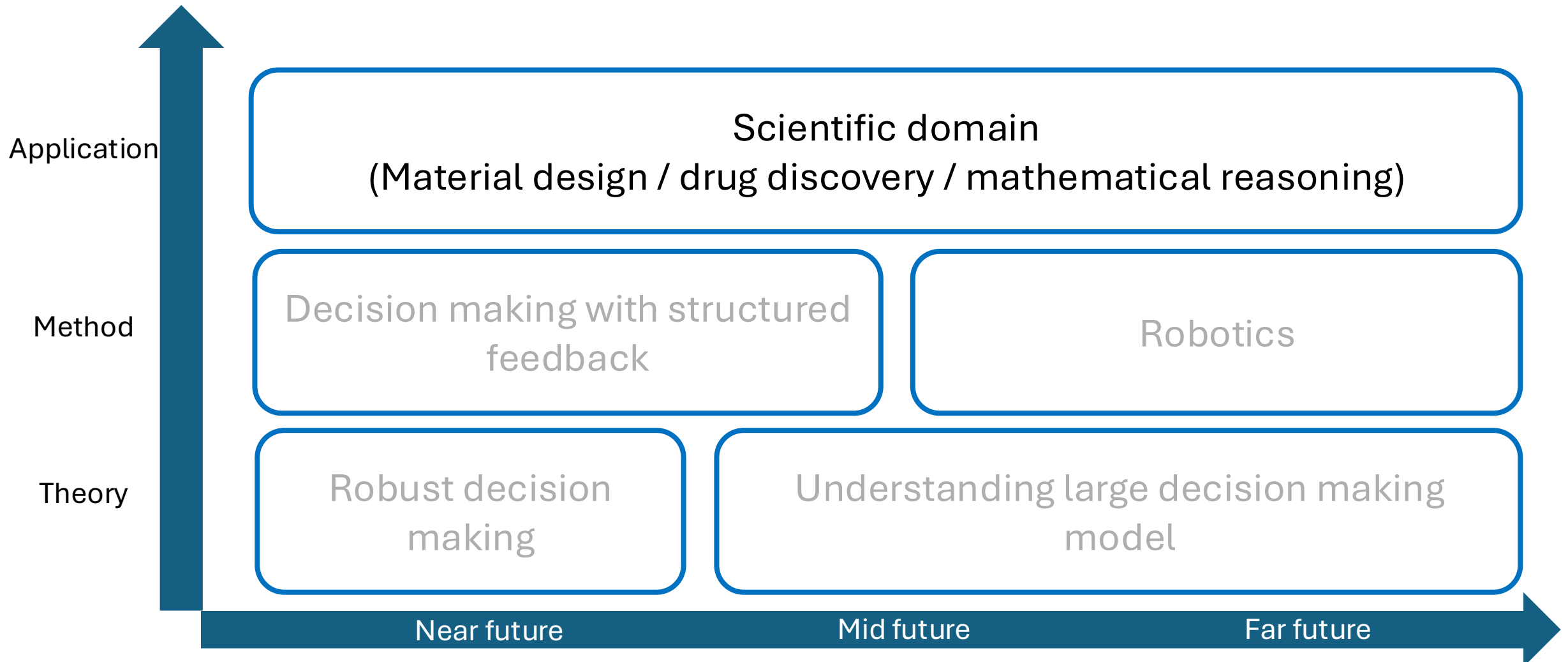


Auto
agriculture



Haircut

Future vision of decision making



Decision making for mathematical reasoning

- Prove Markov's inequality / Goldbach's conjecture.

CL Prove the Markov's inequality.

Markov's inequality is a fundamental inequality in probability theory that relates the probability of a non-negative random variable being greater than or equal to a certain value to the expected value of the random variable. It states that:

For any non-negative random variable X and any positive number a , the probability that X is greater than or equal to a is no greater than the expected value of X divided by a :

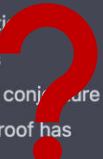
$$P(X \geq a) \leq E[X] / a$$

To prove Markov's inequality, we first note that for any non-negative random variable X and any positive number a , we have:



CL Prove the Goldbach's conjecture.

The Goldbach's conjecture is a well-known unsolved problem in number theory that states that every even integer greater than 2 can be expressed as the sum of two prime numbers. Despite numerous efforts over the centuries, no one has been able to prove this conjecture, and it remains one of the most famous open problems in mathematics. Therefore, as an AI language model, I cannot provide a proof of the Goldbach's conjecture. However, I can tell you that while many mathematicians believe the conjecture is true based on extensive numerical evidence and related results, a rigorous proof has not yet been found.



- Why?
 - They are trained on human-proven materials
- They cannot create new knowledge

Real impact:
It will push the boundary of
human knowledge

Announcement

- Teaching evaluation
 - Opens today!
 - Only 1 student submitted as of this afternoon ...
 - Let's go **above 60%** response rate so that **all** students receive **2 participation points!**
- Final presentations next Monday and Wednesday
 - Monday: Group 1-5, Wednesday: Group 6-10
 - **10 min length** is strictly enforced to ensure all groups can present
 - Followed with my questions
 - I'll give final **2 min** reminders